

Supplementary Information

Distilling solvation physics for simulation-free hydration free energies

Gal Oren and Michael Levitt

This document contains Supplementary Methods, Notes and Tables only. Extended Data figures and their legends are supplied separately with the main manuscript, following Nature’s submission structure. Every numerical table is generated from frozen machine-readable artifacts by `scripts/make_tables.py`. The complete experiment ledger, molecule-level predictions and fold assignments are provided as Supplementary Data rather than compressed into the PDF.

Supplementary Methods

S1. Data-source processing

ARROW reference. The benchmark table was reconstructed from the published ARROW artifacts and contains the 85 neutral water-solvation rows used for the classical and PIMD8 comparison. Molecular names were resolved to structures and canonicalized with RDKit 2026.03.5. The table preserves the source repository, file and publication, experimental uncertainty where available, classical ARROW, PIMD4, PIMD8, family and scaffold fields. No row was removed on the basis of model error.

Global exclusion rule. The benchmark identity set was represented by canonical isomeric SMILES, full InChIKey and the first 14-character connectivity block. Every supervised source used by the final model was canonicalized independently. Any source row sharing a benchmark connectivity was excluded before target aggregation, data splitting or surrogate fitting. Connectivity exclusion deliberately treats stereochemical and tautomeric aliases conservatively. Invalid, disconnected, charged and unresolved structures were removed under source-specific rules. The final release audit repeats these operations from the distributed tables.

CombiSolv-QM water. The water subset of the Chemical Engineering Journal supplementary table was canonicalized and restricted to resolved neutral solutes. Twenty-five benchmark connectivities were removed, leaving 3,963 unique rows from 3,988 source rows. The target is the COSMOtherm water solvation free energy in kcal mol⁻¹. The source has publisher supplementary terms without a standalone redistribution licence; the release records its DOI, URL and processed hash but does not redistribute the derived table.

SoluteML Abraham. The Zenodo collection associated with Chung *et al.*¹ supplied E, S, A, B and L Abraham solute axes. After canonicalization, supported-element filtering and removal of 84 benchmark identities, 8,098 structures remained. All five axes were learned from the same structure matrix.

OpenFF alchemical response. The OpenFF 2.3.0 FreeSolv absolute-solvation archive contained 603 source structures. Eighty-three benchmark connectivities were excluded, leaving 520 structures. The final prior is the sum of two independently predicted targets: the OpenFF 2.3.0 hydration free energy and its experimental residual. This correction is a learned training-time response coordinate; no OpenFF calculation runs at inference.

GBn2 and learned implicit response. The public GNNImplicitSolvent resource supplied 550 benchmark-disjoint structures with GBn2 alchemical and learned implicit-solvent estimates.² The selected coordinate adds predicted GBn2 free energy to a predicted experimental residual. The source contributes no benchmark identity.

MolSolv SMD(water). The archive contains 1,729,545 valid M06-2X/6-31G* SMD(water) conformer calculations.³ After resolving neutral, supported molecules with at most 18 heavy atoms, all eligible structures with at most ten heavy atoms were retained; structures with 11–18 heavy atoms were retained when a SHA-256 rule selected one of eight. This target-independent rule produced 350,391 training structures (348,587 unique connectivities). Eighty-two exact benchmark structures and three connectivity aliases were removed before selection. The target is the reported SMD(water) free energy in kcal mol⁻¹.

ConfSolv water. The H₂O partition of ConfSolv contains 5,392,567 conformer records for 43,693 raw solutes, generated with TPSS-D3(BJ)/def2-TZVP, GFN2-xTB RRHO and COSMO-RS BP-TZVP.⁴ Energies were converted from kJ mol⁻¹ to kcal mol⁻¹. After neutral-structure resolution, deduplication, chemistry filtering and removal of 13 benchmark connectivities, 39,878 connectivities remained. Conformer weights were used to calculate gas, solution and hydration conformer corrections and distributional solvent-response summaries.

Exact source versions, licences, hashes and reproduction commands are listed in Supplementary Table 2 and `repro/DATA_PROVENANCE.md`.

S2. Response-target definitions

SolvAI uses exactly 15 predicted priors (Supplementary Table 1): one CombiSolv-QM/COSMOtherm response, five Abraham axes, one corrected OpenFF response, one corrected GbN2 response, one SMD(water) response and six ConfSolv summaries. All energy-like targets are represented in kcal mol⁻¹. Abraham axes retain the source scale. Corrected physical responses are sums of two independently fitted structure-surrogate outputs, not corrections using a query experimental value.

For a molecule x , let $\hat{r}_j(x)$ denote response surrogate j . The endpoint input is

$$\phi(x) = [d_1(x), \dots, d_{2265}(x), \hat{r}_1(x), \dots, \hat{r}_{15}(x)], \quad (1)$$

where $d(x)$ contains 217 RDKit descriptors and a 2,048-bit radius-2 Morgan fingerprint. This definition is identical in evaluation and deployment.

S3. Teacher models

D-MPNN teachers. The CombiSolv-QM and MolSolv teachers use Chemprop 2.2.4. Both begin from CHEMELEON molecular message-passing weights. The frozen checkpoints contain six message-passing steps, 2,048-dimensional hidden states, mean aggregation, ReLU activation and no message-passing dropout or batch normalization. The regression head has two 256-unit hidden layers and dropout 0.1. The CombiSolv model also receives Chemprop RDKit-2D features; the MolSolv model receives the molecular graph alone. Both use an initial learning rate of 3×10^{-5} , maximum learning rate of 10^{-4} and final learning rate of 10^{-5} with PyTorch seed 20260826. CombiSolv used batch size 64, 30-epoch cap and patience 7; MolSolv used batch size 256, 25-epoch cap and patience 6. Source splits were deterministic: 80/10/10 for CombiSolv and 90/5/5 for MolSolv.

ExtraTrees teachers. All tree teachers receive the frozen RDKit/Morgan structure matrix. Abraham models use 160 estimators, maximum feature fraction 0.7, minimum leaf size 2 and target-specific seeds beginning at 20260826. OpenFF and implicit-solvent models use 500 estimators and maximum feature fraction 0.7. The final-value heads use minimum leaf size 1; residual heads use minimum leaf size 2. Other scikit-learn 1.7.2 defaults are retained: squared-error criterion, no bootstrap, unlimited depth and minimum split size 2. Target-specific seeds increment by 1,009.

ConfSolv teachers. LightGBM 4.7.0 regressors receive 217 RDKit descriptors and the 256 highest-variance radius-2 Morgan bits in the ConfSolv source. Each final artifact uses 180 estimators, L1 regression objective, learning rate 0.04, 31 leaves, minimum child size 20, row subsample 0.8, column fraction 0.55 and L_2 regularization 0.1. Eleven physical targets were fitted for the reusable artifact; the six listed in Supplementary Table 1 enter SolvAI. The final models use all finite source labels after a fixed 90/10 validation screen.

S4. Endpoint training data

The endpoint external pool begins with a benchmark-disjoint, connectivity-resolved merge of FreeSolv and the 8,780-pair CombiSolv-Exp collection.^{5,6} The legacy table contains 1,147 rows: 69 FreeSolv-only identities, 588 CombiSolv-Exp-only identities and 490 identities present in both sources, resolved to one connectivity-level record. SoluteML dGsolvDB3 added 3,928 new benchmark-disjoint identities.¹ The frozen endpoint mask retained all 1,147 legacy rows and 133 SoluteML-only rows supported by at least two source measurements, giving 1,280 external experimental labels. Source groups and rules appear in Supplementary Table 3. The derived merged endpoint table is not redistributed because it contains publisher-supplement records without a standalone redistribution licence; deterministic reconstruction is documented.

Within an outer fold, the 1,280 external rows have weight 1 and the available ARROW outer-training molecules have weight 3. The held-out ARROW experimental label has weight zero because it is absent. The teacher outputs for public endpoint rows are calculated by the same frozen structure surrogates as the benchmark rows. No PIMD8 or PIMD2 value is included in the selected endpoint features.

S5. Endpoint model

Each endpoint member is a scikit-learn 1.7.2 `ExtraTreesRegressor` with 360 estimators, maximum feature fraction 0.7, minimum leaf size 2, squared-error criterion, no bootstrap and unlimited depth. Seeds are 11, 29 and 47. The response vector is concatenated to the structure matrix without target-dependent feature scaling. Predictions are the arithmetic mean of the three members; their sample standard deviation is available as a model-disagreement diagnostic but is not used to route molecules or invoke simulation.

The public deployment model was refitted, after all evaluation was frozen, on the 1,280 external labels and all 85 reference experimental labels using the same configuration. Its purpose is prediction for new molecules. Every manuscript accuracy value instead comes from strict OOF models in which the corresponding reference label was unavailable.

S6. Validation partitions

The primary fixed result uses five frozen random folds. The response block was chosen during the completed campaign and evaluated without fitting the held-out endpoint label. The selection-adjusted analysis performs feature-block selection inside each outer fold: the outer-training set alone generates inner-OOF predictions for the narrow, SMD and ConfSolv additions, then selects a block before predicting the untouched outer fold.

Five complete repeat partitions use seeds 314159, 271828, 161803, 141421 and 173205. Both the fixed model and nested selection were rerun for every partition; no repeat was discarded. Family GroupKfold confines each functional-group family to one held-out group. Scaffold validation uses the frozen scaffold annotations and keeps identical scaffold membership together. Exact molecule-level assignments are Supplementary Data 3; fixed and nested repeat values are Supplementary Table 4.

S7. Statistical analysis

MAE is defined as $n^{-1} \sum_i |\hat{G}_i - G_i|$ and is the primary endpoint. RMSE is retained as a secondary descriptive statistic. Fixed and nested intervals use 100,000 molecule-level bootstrap resamples; the fixed seed is 20260827 and the nested seed is 20260828. Paired method comparisons resample per-molecule absolute-error differences, maintaining the pairing of predictions on the same molecular target. Across split repeats we report the arithmetic mean and sample standard deviation. Family results always display n ; no hypothesis test is applied to rank small families.

S8. Lambda-response probes

The predefined short-probe campaign attempted 76 reference molecules at each of $\lambda = 0.1, 0.5$ and 0.9 . Successful counts were 72, 74 and 73; 72 molecules have all three points. Each probe used two path-integral beads, one coupling state and 5 ps. These observables were treated as privileged response fingerprints, not converged free energies. Exported targets include total ligand-solvent $dH/d\lambda$, Coulomb and van der Waals components. The polarization column was identically zero and supplied no learning signal.

For every held-out molecule, response heads were trained only from response labels in the outer-training partition. The matched experiment compared (A) the SMD+ConfSolv endpoint baseline, (B2) predicted three-point PIMD2 response, (B1) a predicted classical-NQE-PIMD hierarchy, (B) both blocks and (C) numerical integration followed by an affine calibration fitted on outer-training molecules. Their MAEs were 0.19592, 0.20136, 0.21474, 0.21901 and 1.51356 kcal mol⁻¹. Response-head MAEs and correlations are distributed in machine-readable form. None of these PIMD-derived coordinates enters final SolvAI.

S9. Artifact verification

The deployment directory contains two D-MPNN checkpoints, four tree-teacher artifacts, one three-member endpoint artifact and a model card. The SHA-256 and byte length of every file are verified against `models/final/manifest.json` (Supplementary Table 6). A smoke test starts from ethanol and benzene SMILES and expects -5.020248 and -0.893576 kcal mol⁻¹, respectively, within 2×10^{-6} kcal mol⁻¹. The audit also searches the runtime dependency graph and feature schema for experimental targets, ARROW values, PIMD values, trajectories, family labels, scaffold labels and fold identifiers.

S10. Runtime benchmarking

The exact release artifact was timed on Linux with Python 3.11.15, an Intel Core i7-4930K CPU and 12 logical CPUs. The installed RTX 3090 was not used. Cold latency was one call in a fresh process. Warm single-molecule latency comprised five calls after one warm-up. Batch throughput comprised three calls with 32 SMILES. Timing includes canonicalization, descriptor generation, D-MPNN teacher processes, tree teachers and endpoint prediction. Peak resident memory used the process high-water mark. Median warm latency was 17.522 s (p95 17.627 s); median

batch time was 17.911 s, or 0.560 s per molecule. The dominant single-query cost is response-model process startup. We did not alter weights or runtime semantics during publication revision.

Published ARROW/PIMD8 requirements are 16–18 h per 1-ns window on one RTX 2080 Ti plus two CPU cores and 15 windows per solute, or 240–270 serial GPU-hours.⁷ SolvAI therefore uses roughly 49,000–55,000 times less total work for an isolated call or 1.54–1.74 million times less at batch-32 amortization. These ratios compare published GPU simulation work with measured CPU inference and are not same-hardware throughput benchmarks.

Supplementary Notes

S1. ARROW identities and FreeSolv

Eighty of the 85 ARROW reference connectivities are present in the current FreeSolv identity table. This establishes molecular overlap, not the historical provenance of every experimental number in ARROW. We therefore do not describe the 85-solute set as an external FreeSolv test and do not report the frozen model over all FreeSolv as an independent benchmark.

S2. Alternative supervision

The experiment ledger preserves candidate representations that did not improve the matched endpoint. Their scientific role is diagnostic: generic physical information was insufficient, compact water-response quantities transferred better than latent embeddings, and sparse high-fidelity response was not predicted accurately enough. These outcomes do not imply that QMPFF, GNNIS, SAPT, OpenFE or the tested three-dimensional encoders are inaccurate for their original tasks.

S3. Simulation-assisted branch

Selective-PIMD and probe-routing analyses were completed before the project objective was frozen to structure-only deployment. They are retained as Extended Data Fig. 4 and in Supplementary Data 1 because they quantify an alternative accuracy–cost frontier. They are explicitly non-deployable under the SolvAI definition and share no code path with the released predictor.

Supplementary Tables

Supplementary Table 1 | The 15 SolvAI response priors

prior	prior_name	physical_meaning	source	model_class	training_size
1	combisolv_qm_teacher	COSMOtherm hydration response	CombiSolv-QM water	CHEMELEON-initialized D-MPNN	3963
2	abraham_e_teacher	excess molar refraction axis E	SoluteML Abraham	ExtraTrees	8098
3	abraham_s_teacher	dipolarity/polarizability axis S	SoluteML Abraham	ExtraTrees	8098
4	abraham_a_teacher	hydrogen-bond acidity axis A	SoluteML Abraham	ExtraTrees	8098
5	abraham_b_teacher	hydrogen-bond basicity axis B	SoluteML Abraham	ExtraTrees	8098
6	abraham_l_teacher	hexadecane-air partition axis L	SoluteML Abraham	ExtraTrees	8098
7	openff_corrected_teacher	explicit-water alchemical hydration response	OpenFF 2.3.0 ASFE	ExtraTrees	520
8	gbn2_corrected_teacher	implicit-solvent hydration response	GBn2 / GNNImplicitSolvent	ExtraTrees	550
9	molsoolv_smd_teacher	SMD(water) solvation response	MolSolv	CHEMELEON-initialized D-MPNN	350391
10	confsolv_gas_conformer_correction_teacher	gas conformer free-energy correction	ConfSolv H2O	LightGBM	39878
11	confsolv_solution_conformer_correction_teacher	solution conformer free-energy correction	ConfSolv H2O	LightGBM	39878
12	confsolv_hydration_conformer_correction_teacher	hydration conformer correction	ConfSolv H2O	LightGBM	39878
13	confsolv_water_gsoolv_std_teacher	dispersion of conformer solvation energies	ConfSolv H2O	LightGBM	39878
14	confsolv_water_response_mean_teacher	mean conformer solvent response	ConfSolv H2O	LightGBM	39878
15	confsolv_water_response_std_teacher	dispersion of conformer solvent response	ConfSolv H2O	LightGBM	39878

Supplementary Table 2 | Teacher-source provenance and filtering

source	scientific_role	original_records	filtered_records	benchmark_overlaps_removed	license
MolSolv SMD(water)	water-specific quantum-continuum response teacher	1729545.0	350391	82	CC-BY-4.0
ConfSolv H2O	water conformer and solvent-response hierarchy	5392567.0	39878	13	CC-BY-4.0
SoluteML Abraham	empirical solute-response axes	—	8098	84	CC-BY-4.0
OpenFE/OpenFF 2.3.0 FreeSolv ASFE	explicit-water alchemical response teacher	603.0	520	83	CC-BY-4.0
GBn2 / GNNImplicitSolvent	implicit and learned explicit-water response teacher	550.0	550	0	CC-BY-SA-4.0 data
CombiSolv-QM water	COSMOtherm water-response teacher	3988.0	3963	25	publisher supplementary terms; no standalone data license

Supplementary Table 3 | Endpoint experimental training sources

source_group	selected_records	selection_rule	endpoint_target
FreeSolv only	69	legacy benchmark-disjoint merge	experimental hydration
CombiSolv-Exp only	588	legacy benchmark-disjoint merge	experimental solvation in water
FreeSolv + CombiSolv-Exp identity	490	one connectivity-level merged record	duplicate-resolved experimental hydration
SoluteML dGsolvDB3	133	at least two source measurements	consensus experimental hydration

Supplementary Table 4 | Fixed and nested repeat values

repeat	split_seed	method	n	mae
0	314159	Nested selection	85	0.20569
0	314159	narrow response + SMD + ConfSolv response	85	0.20737
1	271828	Nested selection	85	0.20667
1	271828	narrow response + SMD + ConfSolv response	85	0.19786
2	161803	Nested selection	85	0.21127
2	161803	narrow response + SMD + ConfSolv response	85	0.20851
3	141421	Nested selection	85	0.20761
3	141421	narrow response + SMD + ConfSolv response	85	0.20595
4	173205	Nested selection	85	0.21174
4	173205	narrow response + SMD + ConfSolv response	85	0.19907

Supplementary Table 5 | Random-OOF chemical-family errors

family	n	mae_kcal_mol
Acids	3	0.257
Alcohols	13	0.180
Aldehydes	3	0.127
Alkanes	13	0.231
Alkenes	5	0.074
Amides	4	0.467
Amines	5	0.107
Aromatics	13	0.299
Esters	4	0.064
Ethers	4	0.267
Heterorings	5	0.122
Ketones	5	0.160
Sulfides	4	0.124
Thiols	4	0.093

Supplementary Table 6 | Final artifact SHA-256 manifest

file	sha256	bytes
MODEL_CARD.md	3f80cffbc246f6bcde1eb104e0b1699b686b462a6b53f0a7cf59bb5821901ba2	3637
abraham_teacher.joblib	8c8a6365e7162dbe6ae445ce4004928ed86b2669d3e7a70d3d054f143ca97554	87055391
combisolv_qm.pt	a02e38a69450f67404909b46c6931ee5ba226460037c8caeadcf34cca3ed0690	37452324
confsolv_teacher.joblib	55a1d8b9ef8758f478c2f79b8dce17a04526a9ac39373382dc5ee44c2e115f25	2297076
head.joblib	1ba1449d65b053d7006dd4b47bd616ecb9ed027875f75ce47b4730b12d368d06	29448366
implicit_teacher.joblib	484d5e0c1efa759c8e7acbfcbdd87e218b93ae239e7f35303fe4cd7e310910f8	36316890
model_card.json	610f23c7aa3429a9b981166c9e7bf10d89b1797a8a1e9b43505543c7b30fdc6a	1268
molsolv_smd.pt	2b03d32e038b615518b1509ebc2a7c98b3aba2a3eb2d6564a6f95fd7aa9a41d8	37226928
openff_teacher.joblib	55b7d31a293dd61fe57f5505e015775d7d86f53dca69980d398843706d58bb9b	48211661

Supplementary Table 7 | Runtime measurements

measurement	value
cold single molecule	17.659 s
warm single molecule median	17.522 s
warm single molecule p95	17.627 s
batch 32 median	17.911 s
batch 32 per molecule	0.560 s
peak resident memory	151.8 MiB

Supplementary Data files

Supplementary Data 1 Complete experiment ledger (CSV and XLSX): hypothesis, source, target, model, inference requirement, evaluation regime, leakage rule, MAE, interpretation and final-model role.

Supplementary Data 2 Molecule-level reference predictions (CSV and XLSX): experimental, ARROW, PIMD8 and every headline OOF structure-only prediction.

Supplementary Data 3 Molecule-level random, repeat, family and scaffold assignments (CSV and XLSX).

Supplementary Data 4 Definitive response-prior definitions and teacher-source manifests (CSV and XLSX).

References

- [1] Yunsie Chung, Florence H. Vermeire, Hua Wu, Paul J. Walker, Michael H. Abraham, and William H. Green. Group contribution and machine learning approaches to predict abraham solute parameters, solvation free energy, and solvation enthalpy. *Journal of Chemical Information and Modeling*, 62:433–446, 2022. doi: 10.1021/acs.jcim.1c01103.
- [2] Paul Katzberger and Sereina Riniker. A general graph neural network based implicit solvation model for organic molecules in water. *Chemical Science*, 2024. doi: 10.1039/D4SC02432J.
- [3] MolSolv contributors. The solvation energy dataset for machine learning model—molsolv, 2022.
- [4] ConfSolv contributors. Dataset for “confsolv: Prediction of solute conformer free energies across a range of solvents”, 2023.
- [5] Guilherme Duarte Ramos Matos, Daisy Y. Kyu, Hannes H. Loeffler, John D. Chodera, Michael R. Shirts, and David L. Mobley. Approaches for calculating solvation free energies and enthalpies demonstrated with an update of the freesolv database. *Journal of Chemical & Engineering Data*, 62:1559–1569, 2017. doi: 10.1021/acs.jced.7b00104.
- [6] Jiahui Yu, Chengwei Zhang, Yingying Cheng, Yun-Fang Yang, Yuan-Bin She, Fengfan Liu, Weike Su, and An Su. Solvbert for solvation free energy and solubility prediction: a demonstration of an nlp model for predicting the properties of molecular complexes. *Digital Discovery*, 2, 2023. doi: 10.1039/D2DD00107A.
- [7] Leonid Pereyaslavets, Ganesh Kamath, Oleg Butin, et al. Accurate determination of solvation free energies of neutral organic compounds from first principles. *Nature Communications*, 13, 2022. doi: 10.1038/s41467-022-28041-0.